

→) Fractional Knapsack :

FractionalKnapsack (items, capacity)

Input :

items → list of items with value and weight

Output : maximum total value

Begin

For each item

ratio = value / weight

sort all items in descending order of ratio

totalValue = 0

For each item in sorted order

if item.weight $\leq$ capacity then

~~totalValue = totalValue + item.value~~

~~capacity = capacity - item.weight~~

else

fraction = capacity / item.weight

totalValue = totalValue + (item.value * fraction)

break

Return total Value

End

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o/p : Enter no. of items : 3

Enter value and weight of item :

1 4 2 5 3 6

Enter knapsack capacity : 5

Maximum value in knapsack : 2.50